Reflection and Traceability Report on Physiotherapy Companion

Team 11, technically functional
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[Reflection is an important component of getting the full benefits from a learning experience. Besides the intrinsic benefits of reflection, this document will be used to help the TAs grade how well your team responded to feedback. Therefore, traceability between Revision 0 and Revision 1 is and important part of the reflection exercise. In addition, several CEAB (Canadian Engineering Accreditation Board) Learning Outcomes (LOs) will be assessed based on your reflections. —TPLT]

1 Changes in Response to Feedback

1.1 SRS and Hazard Analysis

The hazard analysis document was improved through the peer review along with TA feedback in several areas for enhancement, including clarifying the roadmap with implementation timelines and rationales for safety requirements (#144), adding justification for future work prioritization (#150), and removing problematic assumptions about user familiarity with smart devices (#136). Inconsistencies between the SRS and Hazard Analysis were resolved (#129), while the system boundaries and components section was expanded with lower-level explanations to remove ambiguity (#135). The FMEA table was further decomposed to represent detailed failure modes rather than general effects (#142), and hazard categorization was clarified for technical hazards that could impact safety (#153). Additional improvements included adding rationales for each critical assumption (#136), addressing inconsistencies between assumptions and failure modes (#143), and noting where mitigation strategies would be expanded (#152). TA feedback further reinforced the need for roadmap rationales, which was addressed alongside similar peer feedback. One comment regarding bibliography completeness (#151) was determined to be invalid as the citation and reference were already properly included.

Source	Issue #	Issue Description	Changes Made	Corresponding Commit
Peer Review	#151	Citation "IppolitoWallace1995" appears but bibliography incomplete	No change - comment invalid as bibliography and in-text citation were complete	None
Peer Review (Group 14)	#144	Unclear description of how/when safety requirements and hazard testing will be implemented	Added section in roadmap outlining rationale for accepting requirements within capstone timeline and rationales for excluded safety requirements	939b502

Source	Issue #	Issue Description	Changes Made	Corresponding Commit
Peer Review (Group 14)	#150	Safety requirements prioritized as ”first” vs ”in the future” without clear justification or timeline	Added clear rationale for future work prioritization	470F490
Peer Review (Group 14)	#136	Missing rationale for each assumption; assumption about user familiarity with smart device should be avoided	Removed assumption about user familiarity; added rationales for each assumption	a103812
Peer Review (Group 14)	#143	Inconsistency between assumption about user familiarity and ”confusing prompts” failure mode	Removed the initial assumption; will update language of ”confusing prompts”	None (different commit)
Peer Review (Group 14)	#129	Inconsistencies between SRS and Hazard Analysis; hazards need prioritization by severity	Updated consistency between SRS and HA	Reference #410 PR
Peer Review (Group 14)	#135	System components need lower-level explanation with sample hazards; authentication module lacks detail	Added changes to HA system boundaries, splitting section to remove ambiguity	4bd898a
Peer Review (Group 14)	#142	FMEA rows should represent detailed failure modes rather than effects	Included further decomposition of table within FMEA	Different commit
Peer Review (Group 14)	#153	Technical hazards could also be safety hazards leading to unsafe exercise	Clarified hazard categorization	4381828
Peer Review (Group 14)	#152	Document lacks detailed mitigation strategies and implementation specifics	Mitigation included; will add section about effectiveness	None (different section)
TA	N/A	For Section 8 (Roadmap), including rationale would be helpful for which safety requirements to implement and why they are in that time frame	Updated the section with timelines and rationale for why each safety requirement was placed in each term	Similar to #144

Table 1: Feedback and changes for SRS and Hazard Analysis

1.2 Design and Design Documentation

1.3 VnV Plan and Report

2 Extras

[Summarize the extras that were tackled by this project. Extras can include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. Extras should have already been approved by the course instructor as included in your problem statement. —TPLT]

3 Design Iteration (LO11 (PrototypeIterate))

[Explain how you arrived at your final design and implementation. How did the design evolve from the first version to the final version? —TPLT]

[Don't just say what you changed, say why you changed it. The needs of the client should be part of the explanation. For example, if you made changes in response to usability testing, explain what the testing found and what changes it led to. —TPLT]

4 Design Decisions (LO12)

[Reflect and justify your design decisions. How did limitations, assumptions, and constraints influence your decisions? Discuss each of these separately. —TPLT]

5 Economic Considerations (LO23)

Yes, there is a clear market for our product. Existing physiotherapy companion apps such as Physiapp and Physiomobile offer progress tracking (checklist-based) and gamification (streaks), but they lack real-time exercise form feedback. Our app fills this gap by using OpenCV and MediaPipe pose landmarks to analyze patient movement during home exercises, giving immediate corrective feedback. Additionally, we provide value back to the physiotherapist by automatically sharing tracked metrics (e.g., reps, form quality) together with subjective patient data (pain levels, perceived exertion). This addresses the gap for patients and physios between appointments.

Marketing Approach

Marketing would target physiotherapy clinics and independent practitioners. Strategies include:

- Direct outreach to private clinics via email and informational webinars.
- Free trial periods for clinics to demonstrate improved patient adherence and outcomes.

- Testimonials and case studies from early-adopter clinics.

Cost to Produce Application

We estimate the cost to develop our application to a production-ready version (iOS/Android app, backend dashboard for physios, pose-analysis pipeline) at:

- Development (4 months, 2-3 engineers): \$60,000 - \$80,000.
- Exercise pose analysis expertise (consultation with physiotherapists to validate exercise models and feedback logic): \$10,000 - \$15,000.
- Cloud infrastructure (AWS/GCP for pose inference): \$5,000 initial setup + \$500/month.
- Legal/regulatory (privacy, medical device classification, liability for recording and providing feedback): \$15,000 - \$20,000
- Marketing launch: \$5,000.

Total initial cost: approximately \$95,000 - \$125,000.

Pricing Model

We would not charge individual patients. Instead, we adopt a subscription model for physiotherapy clinics, tiered by the number of active patients per month. A "small private clinic" typically has 2-3 physiotherapists, each seeing 25-50 patients per week. Therefore, reasonable patient tiers are:

- **Tier 1 (up to 50 active patients / month):** \$50 / month
- **Tier 2 (up to 150 active patients / month):** \$100 / month
- **Tier 3 (unlimited patients):** \$200 / month

Breakeven Analysis

Assume an average subscription of \$100 / month per clinic (weighted toward Tier 2). Monthly recurring costs (cloud, support) are roughly \$500 for every 100 clinics. Development cost is a one-time \$110,000.

Monthly profit per clinic after variable costs: \$100 - \$5 = \$95. Number of clinic-months needed to recover development cost:

$$\frac{\$110,000}{\$95 \text{ per clinic-month}} \approx 1,158 \text{ clinic-months.}$$

If we acquire 100 clinics in the first year, breakeven occurs after about 11.6 months. In terms of total clinics, we would need roughly 97 clinics subscribing for one full year to break even.

Potential Users (TAM / SOM)

- **Total Addressable Market (TAM):** All physiotherapists in Canada - approximately 22,000 registered physiotherapists (Canadian Institute for Health Information). Assuming an average of 5 physiotherapists per private clinic (typical for a medium-sized clinic), that gives roughly 4,400 clinics.

- **Serviceable Available Market (SAM):** Private clinics in Ontario, where our initial launch would focus. Ontario has roughly 6,000 physiotherapists, translating to about 1,200 private clinics.

- **Serviceable Obtainable Market (SOM):** In the first two years, with targeted marketing, we could realistically capture 1% of the Ontario market - about 12 clinics. That would generate monthly revenue of $1,200 (at $100 average). With successful promotion and marketing of the application, the application could grow to 5% (60 clinics) after three years, achieving monthly revenue of $6,000 and approaching a breakeven point.

6 Reflection on Project Management (LO24)

[This question focuses on processes and tools used for project management. —TPLT]

6.1 How Does Your Project Management Compare to Your Development Plan

[Did you follow your Development plan, with respect to the team meeting plan, team communication plan, team member roles and workflow plan. Did you use the technology you planned on using? —TPLT]

6.2 What Went Well?

[What went well for your project management in terms of processes and technology? —TPLT]

6.3 What Went Wrong?

[What went wrong in terms of processes and technology? —TPLT]

6.4 What Would you Do Differently Next Time?

[What will you do differently for your next project? —TPLT]

7 Ethics and Equity (LO18)

[Describe how your team applied the principle of equity and universal design to ensure equitable treatment of all stakeholders. —TPLT]

8 Reflection on Capstone

[This question focuses on what you learned during the course of the capstone project. —TPLT]

8.1 Which Courses Were Relevant

[Which of the courses you have taken were relevant for the capstone project? —TPLT]

Matthew

Throughout this project, many courses were relevant to our capstone project. More specifically for the documentation, several of my communication and documentation related courses touched on how to effectively communicate the software specifications. For example, SFWRENG 3A04 - Large System Design, where I had to follow and write software specification. Furthermore, with the coding aspect, I applied learning from, SFWRENG 3S03 - Software Testing, to help write the unit tests. SFWRENG 4HC3 - Human Computer Interfaces, helped with how to design with human compatibility as well as how to conduct usability tests for our application effectively. Finally, the design of databases uses practices from SFWRENG 3DB3 - Databases and SFWRENG 2OP3 - Object Oriented Programming was used to help for the design of the backend Python code. Overall, the application of the courses helped us throughout the capstone project.

8.2 Knowledge/Skills Outside of Courses

[What skills/knowledge did you need to acquire for your capstone project that was outside of the courses you took? —TPLT]

Matthew

Throughout this project, I needed to learn skills outside of my course work. Some of the skills that I needed to learn was working with computer vision and how the pose detection worked. In order to learn these skills, I had to look through the documentation as well as work through tutorials to then try and apply it to the problem of our project. Furthermore, when working on some of the documents for our project, I had to refer to previous course work but also how it is written.